Mugilan Mariappan

PhD Candidate in Computer Science
Systems & Parallel Computing | SFU

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Research Interests

High-performance Graph Processing, Streaming Graph Systems, Graph Neural Network (GNN) Infrastructure, and Graph Database Systems.

Education

Ph.D. in Computing Science — Burnaby, Canada
Simon Fraser University — 2018 – 2026 (expected)
Dissertation: Dependency-Driven Incremental Processing of Streaming Graphs
Advisor: Dr. Keval Vora | **Expected Defense:** October 2026

B.E. in Computer Science — Coimbatore, India
Coimbatore Institute of Technology (GPA: 8.5/10) — 2010 – 2014

Publications

Citation counts from Google Scholar, August 2026.

1. **GraphBolt: Dependency-Driven Synchronous Processing of Streaming Graphs**
Mugilan Mariappan and Keval Vora.
European Conference on Computer Systems (EuroSys), 2019. | 179 citations

2. **DZiG: Sparsity-Aware Incremental Processing of Streaming Graphs**
Mugilan Mariappan, Joanna Che, and Keval Vora.
European Conference on Computer Systems (EuroSys), 2021. | 79 citations

3. **Anti-Vertex for Neighborhood Constraints in Subgraph Queries**
Kasra Jamshidi, **Mugilan Mariappan**, and Keval Vora.
ACM SIGMOD Workshop on GRADES-NDA, 2022. | 7 citations

4. **Threshold Strategy for Leaking Corner-Free Hamilton-Jacobi Reachability with Decomposed Computations**
Chong He, **Mugilan Mariappan**, Keval Vora, and Mo Chen.
IEEE Conference on Decision and Control (CDC), 2025. | 1 citation

Research Software

- **GraphBolt/DZiG**: Open-source C++ streaming graph engine (MIT) and the reference implementation for BSP-guaranteed incremental graph processing. **150+ stars, 25 forks** on GitHub.

Research Experience

Simon Fraser University — Burnaby, Canada
Graduate Research Assistant, PDC Lab (PDCL) — 2018 – Present

- Built **GraphBolt**, a dependency-driven streaming graph engine that refines aggregated dependencies to **guarantee BSP semantics** under continuous mutation, generalized across a broad class of iterative graph algorithms. Achieves a **median 18× speedup (up to 232×)** over recomputation from scratch across five algorithms on graphs up to **6.6B edges**.

- Built **DZiG**, a sparsity-aware incremental engine that treats value stabilization as a first-class construct, pruning redundant computation at every stage of refinement. **Pushes dependency-driven processing past 10M simultaneous edge mutations**, orders of magnitude beyond prior systems. Incorporates a linear-regression-based adaptive scheduler that improves performance at the highest mutation rates.
- Formalized matching semantics and prototyped grammar extensions to **Cypher** for expressing negative neighborhood constraints in subgraph queries, enabling automatic reasoning and optimization during query evaluation.

LinkedIn Graph Systems Lab — Sunnyvale, USA
Graduate Research Intern — Sep 2022 – Dec 2022

- Benchmarked LinkedIn's proprietary **Liquid Graph Database** against the complex-query workloads of the **LDBC Social Network Benchmark**, diagnosing the root causes of slow query execution and identifying optimizations achievable within the existing engine.
- Surfaced critical design constraints for graph query processing with mentors at LinkedIn and **Microsoft Research (Gray Systems Lab)**, informing a rearchitecture of Liquid's query processing module undertaken over the following two years.

Industry Experience

athenaHealth — Chennai, India
Software Engineer — May 2014 – July 2017

- Rearchitected the **athenaText** secure messaging module, reducing application crashes by **50%** and eliminating UI stalls during message sync and refresh.
- Maintained and extended the indexing and ranking components of search for **Epocrates**, a clinical reference application used by over 50% of US physicians.

Teaching & Mentoring

Teaching Assistant (SFU): Distributed Systems (CMPT 431), Operating Systems (CMPT 300).
Mentorship:

- **Graduate Mentor:** Guided 7+ students in systems research and performance engineering at PDCL.
- **Key Collaborations:** Mentored Joanna Che (M.Sc.) for the **DZiG** project and supervised undergraduate researchers including M. Morrison, A. Scridon, and N. Kuwayama on graph infrastructure.

Honors & Awards

Scholarships: Helmut & Hugo Eppich Family Graduate Scholarship (2022); Special Graduate Entrance Scholarship (2018).
Fellowships: Graduate Fellowship (2018, '20, '22, '23); CMPT Graduate Fellowship (2019, '21, '22).
Awards: Best Poster Award (with Joanna Che), SFU CS Research Day 2019.

Professional Service

- **Reviewer:** IEEE Transactions on Knowledge and Data Engineering (TKDE), 2026.
- **External Reviewer:** EuroSys (2020, '24, '27); USENIX ATC (2020, '24); SOSP (2023); SIGMOD (2023); ICDCS (2023); PACT (2020); IEEE TPDS (2023); IEEE TKDE (2022).